

## MTH203: Multivariate Calculus

Monsoon 2025

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Name: \_\_\_\_\_

Roll No.: \_\_\_\_\_

### Tutorial Quiz 2

#### Question:

Consider the functions

$$f(x, y) = y - x$$

$$g(x, y) = x^2 + 2xy$$

1. Find the gradient of the function  $f(x, y)$  at the point  $(2, 1)$ . **1**
2. Find the local maxiam, local minima, and saddle point of the function  $g(x, y)$ . **2**

Please note: There is no partial marking.